

System Name: IPEC

System Type: Enterprise System

Architecture Style: Monolithic MVC

1. System Selection & Problem Definition

1.1 System Type Selected

Enterprise System - A web-based e-commerce platform designed to connect indigenous artisans like B'laan, T'boli, and Maguindanaon residing in General Santos City with local buyers within the city, while preserving cultural heritage through digital commerce. The platform focuses exclusively on local transactions within General Santos City for streamlined operations and community-centered commerce.

1.2 Real-World Problem Identification

Indigenous artisans like B'laan, T'boli, and Maguindanaon communities living in General Santos City face significant challenges in selling their traditional crafts within their own city. Despite creating valuable cultural products like T'nalak weaves, brass accessories, beadwork, and traditional garments, these artisans struggle to connect with local buyers who appreciate and want to purchase authentic indigenous crafts.

Problem Area	Description
Limited Local Visibility	Artisans can only sell in limited venues like Local Market and LGU Small Shop where people won't notice it fast
Small Centralized Marketplace	LGU provides Small Shop like the Blaان Livelihood Center there is no Centralized Shops for

Cultural Disconnection	Many GenSan residents are unaware of the rich indigenous heritage within their own city and the cultural significance of local crafts
Inconvenient Transactions	Buyers and artisans have difficulty coordinating meetups, payments, and deliveries within the city
No Digital Records	Artisans have no documentation of their own products, sales history, or customer base
Authenticity Concerns	Local buyers cannot distinguish authentic handmade crafts from mass-produced imitations sold in malls and tourist shops

This leads to:

- Indigenous artisans in GenSan earning below fair value for their work
- Local buyers purchasing imitation products unknowingly
- Declining practice of traditional crafts among indigenous communities in the city
- Lost opportunity for cultural appreciation
- Difficulty for artisans to build a sustainable livelihood from their craft
- Lack of awareness among GenSan residents about local indigenous culture especially Gen Z

1.3 Proposed Software Solution

Solution: IPEC Platform

IPEC is an enterprise-level e-commerce platform that serves as a digital marketplace exclusively for authenticated indigenous crafts made by indigenous artisans residing in

General Santos City. The platform connects these artisans directly with buyers within the city, and enabling local pickup

The platform provides:

- Direct artisan-to-buyer sales channel within General Santos City
- pickup options
- Cultural story preservation for each product
- Authentication and verification system for artisans and products
- Bilingual interface (Filipino and English)

Solution Approach

Problem	IPEC Solution
Limited Local Visibility	Centralized online marketplace for GenSan buyers to discover local indigenous artisans
Small Centralized Marketplace	Dedicated platform exclusively for indigenous crafts in General Santos City
Cultural Disconnection	Cultural archive featuring stories, techniques, and heritage of each indigenous group in GenSan
Inconvenient Transactions	In-app coordination for local pickup points
No Digital Records	Digital product catalog, sales history, and customer management for artisans
Authenticity Concerns	Artisan verification through CMO - Muslim Concerns and Affairs Division

1.4 Core Purpose

"To create a trusted digital marketplace that connects indigenous artisans in General Santos City directly with local buyers, ensuring fair compensation for authentic crafts while fostering cultural appreciation and heritage preservation within the GenSan community."

In simple terms: *"Pag-ugnay sa mga katutubong manggagawa at mga mamimili sa General Santos City para sa patas na kalakalan at pagpapahalaga sa ating kultura."*

(Connecting indigenous artisans and buyers in General Santos City for fair trade and cultural appreciation.)

1.5 Primary Functionalities

Functionality	Description	Priority
Artisan Registration & Verification	Registration with GenSan residency verification or barangay certification	Must Have
Product Listing Management	Add/edit products with photos, cultural stories, materials, pricing; product categories by craft type and tribe	Must Have
Shopping Cart & Checkout	Multi-item cart, order summary	Must Have
Local Payment Processing	GCash, Maya, cash on pickup	Must Have
Pickup Options	designated pickup points, artisan location pickup	Must Have
Order Management	Order tracking, status updates, pickup coordination	Must Have

Admin Dashboard	Platform management, artisan approval, order monitoring, dispute resolution, basic analytics	Must Have
Artisan Dashboard	Sales summary, product management, order notifications, earnings tracking, payout requests	Must Have
Cultural Heritage Section	Profiles of IP communities in GenSan, craft documentation, artisan stories	Should Have
Review & Rating System	Buyer reviews, artisan ratings	Should Have
Search & Filter	Search by product name, category, tribe, price range, filter and sort options	Should Have
Notifications	Order updates via email; new product alerts for buyers	Should Have
Bilingual Interface	Filipino (Tagalog) and English language options	Should Have
Artisan Analytics	Basic sales reports, popular products, monthly earnings summary	Could Have
Promotional Features	Festival Celebration, Discounts	Could Have
In-App Messaging	Buyer-artisan communication for order inquiries and customization requests	Could Have

2. Define Objectives, Scope & Users

2.1 SMART Objectives

Objective 1: Artisan Onboarding in General Santos City

- **Specific:** Register and verify at least 30 indigenous artisans from B'laan, T'boli, and Maguindanaon communities residing in General Santos City.
- **Measurable:** 30 verified artisan accounts with complete profiles, valid GenSan residency, tribal affiliation documentation, and at least 2 products listed each
- **Achievable:** Through partnerships with CMO - Muslim Concerns and Affairs Division.
- **Relevant:** Building a base of verified local artisans ensures product authenticity and enables sustainable local operations
- **Time-bound:** Within 3 months after platform launch

Objective 2: Local Product Catalog Development

- **Specific:** Build a catalog of at least 30 authenticated indigenous craft products made by GenSan-based artisans, each with complete documentation including cultural significance
- **Measurable:** 50 products with photos, descriptions, pricing, cultural stories, and authentication codes
- **Achievable:** With 30 artisans listing an average of 3-4 products each
- **Relevant:** A diverse catalog of local products attracts GenSan buyers and showcases the variety of indigenous crafts available within the city
- **Time-bound:** Within 3 months after platform launch

Objective 3: Local Buyer Registration

- **Specific:** Register at least 30 active buyer accounts from General Santos City with verified local addresses
- **Measurable:** 30 buyer accounts with complete profiles, verified GenSan addresses, and at least one completed transaction
- **Achievable:** Through local digital marketing (Facebook groups specific to GenSan), partnerships with local schools, offices, and malls, and word-of-mouth promotion
- **Relevant:** A strong local buyer base ensures demand for artisan products and platform sustainability
- **Time-bound:** Within 3 months after platform launch

Objective 4: Successful Local Transaction Completion

- **Specific:** Achieve a transaction success rate of at least 75% for all orders within General Santos City, from checkout to successful pickup
- **Measurable:** 75% of confirmed orders are successfully delivered or picked up within the city
- **Achievable:** Through reliable local convenient pickup points
- **Relevant:** High transaction success builds trust among both artisans and buyers, encouraging repeat transactions
- **Time-bound:** Maintain this rate starting from Month 2 of operations

Objective 5: Artisan Income Generation

- **Specific:** Enable participating artisans to earn at least ₱3,000 per month in additional income through platform sales
- **Measurable:** Average monthly earnings of ₱3,000 per active artisan, tracked through platform sales data
- **Achievable:** By providing direct access to local buyers, eliminating middlemen, and promoting artisan products through the platform
- **Relevant:** Directly addresses the core problem of unfair compensation and makes traditional crafting a viable livelihood
- **Time-bound:** Within 5 months of active artisan participation

2.2 Project Scope

Included Features

1. Platform and Technology

- Web-based enterprise application accessible via modern browsers (Chrome, Firefox, Safari, Edge)
- Responsive design optimized for desktop, tablet, and mobile viewing
- **Monolithic MVC Architecture:**
 - **Presentation Layer:** Laravel Blade templates (server-side)
 - **Business Logic Layer:** Laravel PHP (controllers, services)
 - **Data Layer:** MySQL database (Eloquent ORM)
- Secure HTTPS connection for all transactions

2. User Account Management

Artisan Accounts:

- Registration with required documents:
 - Valid ID with General Santos City address
 - Barangay certificate or residency proof
 - Tribal affiliation proof (CMO - Muslim Concerns and Affairs Division)
- Profile management:
 - Personal information (name, contact number, address within GenSan)
 - Artisan bio and story
 - Profile photo
 - Tribal affiliation (B'laan, T'boli, or Maguindanaon)
 - Craft specialization
- Product management dashboard
- Order management and notifications
- Sales and earnings tracking

Buyer Accounts:

- Registration via email
- Profile management:
 - Name and contact information
 - Preferred pickup locations
- Order history and tracking
- Review and rating submission

Admin Accounts:

- Full platform access
- Artisan verification and approval
- Financial tracking and payout processing
- Content management
- Report generation

3. Product Management

Product Listing Features:

- Multiple photos per product (up to 3)
- Product name and detailed description
- Cultural story and significance
- Materials used
- Techniques/methods used in creation
- Dimensions and weight
- Pricing (artisan sets the price)
- Stock quantity
- Estimated creation/preparation time
- Tribal origin (B'laan, T'boli, or Maguindanaon)

Product Categories:

- Textiles and Weaves
 - T'nalak (T'boli abaca weave)
 - Malong (Maguindanaon tube garment)

- Traditional fabrics
- Accessories
 - Beadwork (necklaces, bracelets, earrings)
 - Brass items (Maguindanaon brassware)
 - Traditional jewelry
- Home Décor
 - Wall hangings
 - Table runners
 - Decorative items
- Traditional Clothing
 - Complete traditional attire
 - Modern clothing with indigenous design
- Bags and Pouches
 - Woven bags
 - Beaded pouches
- Others
 - Musical instruments
 - Ceremonial items
 - Custom/commissioned pieces

Product Authentication:

- Unique product code for each item
- Verified artisan badge displayed on product

- Tribal origin label

4. E-Commerce Functionality

Product Browsing and Search:

- Browse by category
- Browse by tribe
- Browse by artisan
- Browse by price range
- Keyword search
- Filter options
- Sort options

Shopping Cart:

- Add/remove items
- Quantity adjustment
- View item details
- Save items for later
- Cart total calculation

Checkout Process:

- Review cart items
- Select pickup option:
 - **Pickup Options:**
 - Artisan location pickup

- Select payment method
- Add order notes (special instructions)
- Order confirmation

Payment Methods:

- **GCash** - Manual payment with reference number verification
- **Maya** - Manual payment with reference number verification
- **Cash on Pickup (COP)** - Pay upon pickup at designated location

5. Order Management

Order Status Flow:

1. Order Placed - Buyer completes checkout
2. Pending Payment - Waiting for payment confirmation for non-COD
3. Payment Confirmed - Payment verified by admin/artisan
4. Processing - Artisan preparing the order
5. Picked Up - Order received by buyer
6. Completed - Transaction finished, review requested
7. Cancelled - Order cancelled (with reason)
8. Refunded - Payment returned (with proof)

Order Features:

- Estimated pickup date
- Pickup coordination between buyer and artisan
- Order notifications (Email)
- Order history for both buyer and artisan

- Order cancellation (with conditions)
- Issue reporting and dispute system

6. Cultural Heritage Section

- Indigenous People Community in General Santos City
 - History and background
 - Traditional practices
 - Crafts and specializations
 - Community locations within GenSan

Craft Documentation:

- Types of crafts per tribe
- Materials and techniques
- Cultural significance and symbolism
- Traditional uses
- Modern adaptations

Artisan Stories:

- Featured artisan profiles
- Artisan journey and experience
- Photos and videos of artisans at work

7. Admin and Management Features

Admin Dashboard:

- Overview statistics:
 - Total registered artisans

- Total registered buyers
- Total products listed
- Total orders (daily, weekly, monthly)
- Total sales revenue
- Pending verifications

Artisan Management:

- Verification queue with document review
- Approve/reject artisan applications
- Artisan profile management
- Performance monitoring
- Deactivation

Product Management:

- Product approval queue
- Featured product selection
- Product removal

Order Management:

- View all orders
- Order status monitoring
- Dispute resolution interface
- Cancellation and refund processing

Financial Management:

- Transaction records

- Artisan earnings tracking
- Payout processing and history
- Basic financial reports

Content Management:

- Homepage banner management
- Featured artisans/products
- Announcements and notifications
- Cultural content updates

Reports:

- Sales reports (daily, weekly, monthly)
- Artisan performance reports
- Product popularity reports
- Buyer activity reports
- Exportable to PDF

8. Communication Features

Notifications:

- Email notifications for:
 - Order confirmation
 - Payment confirmation
 - Order status updates
 - New messages

In-App Messaging:

- Buyer-to-artisan messaging for:
 - Product inquiries
 - Customization requests
 - Order coordination
- Message history

9. Language Support

- Filipino (Tagalog) as primary interface language
- English as secondary interface option
- Indigenous language terms preserved in:
 - Product names
 - Cultural descriptions
 - Craft terminology
- Glossary of indigenous terms with explanations

Excluded Features

1. Geographic Expansion

- Buyer addresses limited to General Santos City barangays only
- Artisan registration limited to GenSan residents only

2. Other Platforms

- Mobile users access through responsive web browser
- No desktop application

3. Advanced Payment Features

- No direct payment gateway API integration (manual verification only)
- No credit card processing
- No PayPal or international payment methods
- No installment payments or buy-now-pay-later
- No in-app wallet system

4. Advanced Logistics

- No own delivery fleet
- No automated courier booking API
- No real-time GPS tracking of deliveries
- No warehouse or fulfillment center
- No packaging provided (artisans handle own packaging)

5. Advanced E-Commerce Features

- No auction or bidding system
- No wholesale ordering
- No bulk discounts
- No subscription or membership system
- No loyalty points or rewards program
- No gift cards or vouchers

6. Marketing Automation

- No automated email marketing
- No social media integration for auto-posting
- No affiliate or referral program

- No influencer management tools
- No paid advertising integration

7. Advanced Technology

- No AI-powered recommendations
- No chatbot support
- No demand forecasting
- No automated pricing suggestions
- No image recognition for products

8. Financial Services

- No microfinancing or loans for artisans
- No insurance for products or shipments
- No tax computation or BIR compliance tools

9. Community Features

- No community forum or discussion boards
- No live streaming sales
- No virtual workshops or classes
- No social networking features between buyers

10. Multi-Vendor Complexity

- No multiple stores per artisan
- No franchise or branch system
- No inventory sync with external platforms (Shopee, Lazada)
- No POS integration

2.3 User Personas

User Persona 1 – Lang Dulay

1. Background

- Lang Dulay is 45 years old
- She lives in Barangay Lagao, General Santos City
- She is a skilled beadwork artisan, learning the craft from her mother since age 10
- She is married with three children, her eldest daughter also makes beadwork
- She speaks B'laan at home, Filipino in the community, and understands basic English
- She moved to General Santos City 15 years ago from the mountains of Sarangani

2. Phone and Technology Use

- She uses an Android smartphone
- She can use basic apps:
 - Facebook (to view posts and chat with family)
 - Messenger (for communication)
 - GCash (for receiving payments and paying bills)
 - YouTube (for entertainment and tutorials)
- She has prepaid mobile data (Globe or Smart)
- Internet is sometimes slow in her area
- She needs help from her daughter for unfamiliar app features
- She prefers Filipino language interfaces

3. Craft and Business Situation

- She creates beaded jewelry:

- Necklaces
- Bracelets
- Earrings
- Beaded accessories
- Traditional B'laan accessories
- Time to make one piece: 2 hours to 2 days depending on complexity
- She currently sells through:
 - Weekly bazaars at SM City GenSan (when available)
 - Orders from friends and neighbors
 - Consignment at a small shop in the city
 - Occasional school and office events
- Selling price: ₱150 to ₱1,500 per piece
- Consignment shop takes 40% commission
- Monthly income from crafts: ₱2,000 to ₱5,000 (irregular)

4. Goals Lang Dulay wants IPEC to help her:

- Sell her beadwork directly to buyers in General Santos City
- Reach more customers beyond her current network
- Receive orders even when there are no bazaars
- Show buyers the cultural meaning behind her designs
- Have a record of her products and sales

5. Problems and Frustrations

- Bazaar slots are limited and competitive; she doesn't always get a space

- The consignment shop takes too much of her earnings
- Buyers don't understand the value of handmade beadwork
- She has no way to show her products to people she doesn't personally know
- Coordinating meetups for direct sales is difficult and time-consuming
- Her income is unpredictable because sales depend on events and luck

User Persona 2 – Shean Dela Cruz

1. Background

- Shean is 26 years old
- She works as a bank teller at a BDO branch in General Santos City
- She finished college (BS Business Administration) at Notre Dame of Dadiangas University
- She is interested in supporting local products and indigenous communities

2. Technology Use

- She uses a mid-range Android smartphone (Samsung Galaxy A series)
- She is comfortable with technology:
 - Active online shopper (Shopee, Lazada, Facebook Marketplace)
 - Uses GCash and Maya daily
 - Active on social media (Facebook, Instagram, TikTok)
- She has postpaid mobile plan with unlimited data
- Stable home WiFi connection
- She prefers apps that are easy to use and visually appealing

3. Buying Interests

- She wants to buy indigenous crafts for:
 - Personal accessories (unique jewelry and bags)
 - Gifts for friends and officemates
 - Home decoration
 - Supporting local artisans
- Her budget: ₱200 to ₱3,000 per item
- She currently buys from:
 - Random Facebook sellers (uncertain quality)
 - Mall bazaars (limited selection)
 - Pasalubong shops (mostly generic items)
- She prefers to buy within GenSan for convenience

4. Goals Christine wants IPEC to help her:

- Find authentic indigenous crafts made by real local artisans
- Know the story and meaning behind each piece she buys
- Support verified indigenous artisans directly
- Share her purchases on social media to promote local artisans
- Have a trusted platform for buying unique, meaningful items

5. Problems and Frustrations

- She can't tell if products on Facebook are authentic or mass-produced
- Mall items are expensive and may not benefit actual artisans
- Bazaars are only occasional and timing doesn't always match her schedule
- She worries about meeting strangers for direct transactions

- She wants to know more about the culture, but sellers often don't have information

2.4 Key Stakeholders

Primary Stakeholders

1. Indigenous Artisans in General Santos City (Main Users - Sellers)

- They are the main beneficiaries and product providers
- Their success determines platform success
- They should be involved through:
 - Community consultations during design phase
 - User testing sessions before launch
 - Regular feedback collection after launch
 - Artisan representative input for platform improvements

2. Local Buyers in General Santos City

- GenSan residents interested in indigenous crafts
- Young professionals and students
- Cultural enthusiasts
- Gift buyers
- They drive demand and platform revenue
- Their satisfaction determines repeat business and referrals

Internal Stakeholders

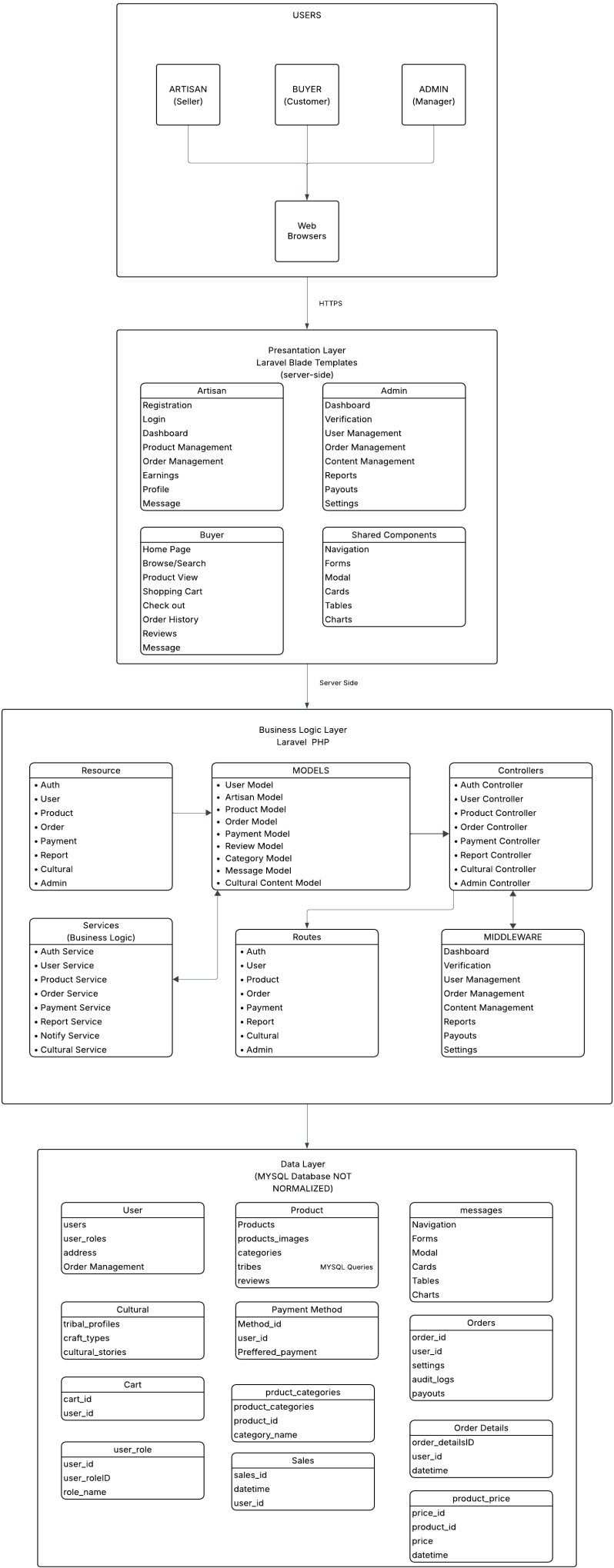
1. Development Team

- Web developers (frontend and backend)
- Database administrator
- Quality assurance tester
- They need:
 - Clear requirements and specifications
 - Understanding of user context and limitations
 - Regular feedback from actual users

2. Course Adviser

- Oversees project timeline, scope, and deliverables
- Coordinates between technical team and stakeholders
- Ensures alignment with objectives
- Manages risks and issues

3. High Level Architecture Diagram



4. Components & Integrations

4.1 Core System Modules

Module	Description	Key Features
1. User Management Module	Handle all user-related operations including registration, authentication, and profile management	Authentication, Registration, Profile Management, Pickup Address Management, Artisan Verification, Role Management
2. Product Management Module	Manage all product-related operations from listing to inventory	Product Listing, Product Images, Category Management, Inventory Management, Product Search, Product Authentication,
3. Order Management Module	Handle the complete order lifecycle from cart to completion	Shopping Cart, Checkout, Order Creation, Order Tracking, Order Status Management, Order Cancellation, Order History
4. Payment Management Module	Handle payment processing, verification, and artisan earnings	Payment Methods (Cash, Cash on Pickup, Gcash, Maya), Payment Submission, Payment Verification, Payment

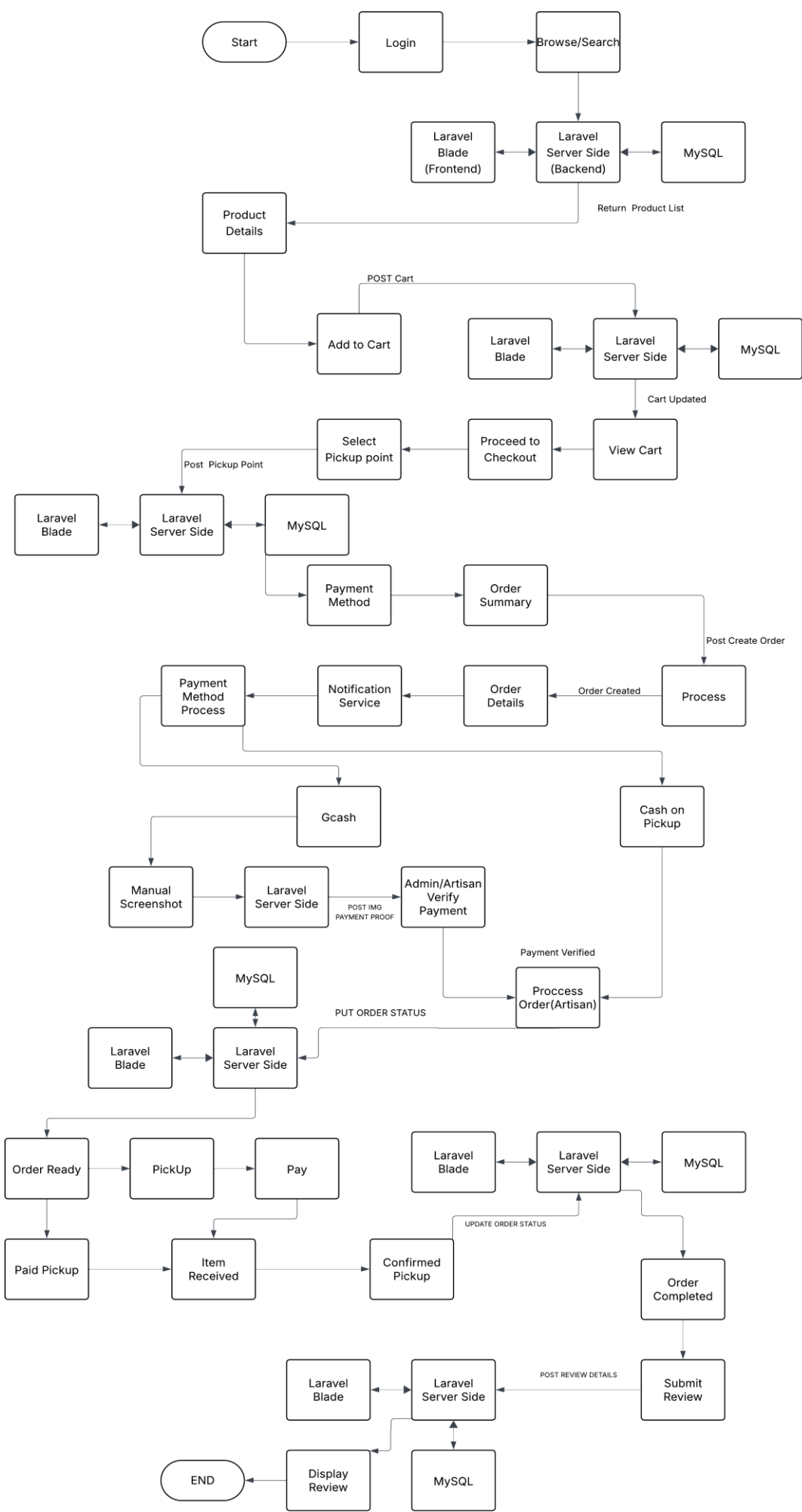
		Status, Artisan Earnings, Transaction History
5. Pickup Management Module	Manage pickup points within General Santos City	Pickup Points, Pickup Coordination, Artisan Location Pickup
6. Cultural Heritage Module	Preserve and showcase indigenous cultural heritage connected to the crafts	Tribal Profiles, Craft Documentation, Artisan Stories, Cultural Glossary, Media Gallery
7. Review & Rating Module	Allow buyers to review products and artisans, building trust in the marketplace	Product Reviews, Review Display, Artisan Ratings, Review Verification, Review Moderation, Review Response
8. Messaging & Notification Module	Enable communication between users and artisans	In-App Messaging, Order Messages, Email Notifications, Push Notifications, Notification Preferences
9. Reporting & Analytics Module	Generate reports and analytics for artisans and administrator	Sales Reports, Artisan Performance, Product Analytics, Buyer Analytics, Financial Reports, Export Functionality
10. Content Management Module	Manage static and dynamic content on the platform	Homepage Management, Announcements, Help

		Center, Cultural Content, FAQ Management
11. Admin Management Module	Provide administrative tools for platform management	Dashboard, User Management, Artisan Verification, Order Management, Payment Management, Content Management, Category Management,

4.2 Third-Party Services / APIs

- 1. **Email Service (SMTP)**- Send transactional emails (order confirmations, notifications, password resets)
- 2. **QR Code Generator** - Generate QR codes for product authentication certificates
- 3. **PDF Generator** - Generate PDF reports, receipts, and certificates
- 4. **File Storage Service** - Store uploaded files (product images, documents, media)
- 5. **Map Service** - Display pickup point locations/artisan locations
- 6. **Real-time chat** – Communication for the artisans and buyers

4.3 Data Flow Diagram



5. Architecture Justification

5.1 Chosen Architecture

Monolithic MVC is a way of building software where everything is in one single application. MVC stands for Model-View-Controller, which separates the code into three parts:

Part	What It Does
Model	Handles data and business rules
View	Shows information to users
Controller	Connects Model and View, processes requests

How This Applies to IPEC:

View (Laravel Blade):

- This is what users see on their screens
- Artisans use it to add their products
- Buyers use it to look at products and buy them
- Administrators use it to manage the whole platform

Controller & Model (Laravel PHP):

- This is the brain of the application

Database (MySQL):

- This is where all information is stored permanently

Laravel Blade for the Front-End (View)

Blade is Laravel's built-in template engine. It helps us create web pages that users see and interact with.

Why we chose it:

Reason	Explanation
Built into Laravel	No need to set up separate tools; everything works together
Simple to learn	Easy syntax that new developers can understand quickly
Fast to develop	We can reuse parts of pages, saving time

Laravel PHP for the Back-End (Controller & Model)

Laravel is one of the most popular PHP frameworks in the world. It helps developers build web applications quickly and securely.

Why we chose it:

Feature	Benefit for IPEC
Easy to read code	Code is clean and understandable
Built-in authentication	User login and registration already included
Eloquent ORM	Easy way to work with database without writing complex queries
Security features	Protection against common attacks (SQL injection, XSS, CSRF)
Affordable hosting	PHP hosting is cheap and widely available in the Philippines

MySQL Database

MySQL is a system that stores and manages data. It is one of the most used databases in the world.

Why we chose it:

Feature	Benefit for IPEC
Reliable	Data is safe and accurate
ACID transactions	When a buyer makes a purchase, all related changes happen together. If one fails, everything is cancelled. This prevents errors.
Works well with Laravel	Laravel and MySQL are designed to work together
Free to use	No need to pay for licenses
Affordable hosting	Many cheap hosting options in the Philippines

5.2 Reasons for Choosing This Architecture

Reason 1: Everything is organized clearly

IPEC has three types of users:

- **Artisans** - need tools to manage products and orders
- **Buyers** - need a smooth shopping experience
- **Administrators** - need to oversee and manage everything

Reason 2: Better security

IPEC handles sensitive information:

- Artisan documents
- Buyer addresses
- Payment records

With our architecture, the database is protected. Users cannot access the database directly. All requests must go through the Controller first. The Controller checks if the request is safe and valid before touching the database.

Reason 3: Easier to develop and maintain

With Monolithic MVC:

- All code is in one place
- Developers can understand the whole system easily
- New team members can learn quickly
- Finding and fixing bugs is simpler

Reason 4: Well-known and well-supported technology

Laravel is a very popular framework with:

- Lots of tutorials and guides are available
- Many developers already know how to use it
- It is easy to find help when we have problems
- The technology is tested and trusted

5.3 Trade-offs We Considered

Trade-off 1: Cost vs. Capability

What We Get	What We Give Up
Low cost - Can run on one server	Limited advanced scaling - May need upgrades if traffic grows very fast
Free tools - Laravel, Blade, MySQL are all free	
Affordable hosting - PHP/MySQL hosting is cheap in Philippines	

Our decision: Cost savings are more important now. We can upgrade later if needed.

Trade-off 2: Simplicity vs. Flexibility

What We Get	What We Give Up
Easy to understand - All code in one place	Limited technology choices - Must use PHP/Laravel for everything
Quick onboarding - New developers learn fast	Cannot use other languages - Even if another language is better for a specific task
Easy debugging - Problems are easier to find	

Our decision: Simplicity is better for a small team with limited time.

Trade-off 3: Performance vs. Development Speed

What We Get	What We Give Up
Fast development - Built-in features save time	Slight overhead - Framework adds small delay compared to raw code
Less code to write - Many features already available	

Our decision: For IPEC's expected size (30 artisans, 200 buyers), the small performance difference is not noticeable. Fast development is more important.

Trade-off 4: Scalability vs. Immediate Needs

What We Get	What We Give Up
Simple setup - No complex scaling from the start	Manual scaling needed - If traffic increases a lot, we must add features manually
Lower initial cost - No need for load balancers now	

Our decision: We designed for future scaling but did not build it yet. No point paying for features we do not need right now.

Trade-off 5: Security vs. Usability

What We Get	What We Give Up
Easy to use - Simple login process	Basic security - No two-factor authentication yet
Accessible - Artisans with limited tech skills can use it	Room for improvement - Can add more security later

Our decision: Many artisans are not very comfortable with technology. Making the platform easy to use is important. Basic security is enough for now, and we can add more later.